

Rank-Based Credit and Strategic Incentives in Open Teams

Q Mechanism Design for Alignment and Control [arXiv:2609.01595](#)

P MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation [arXiv:2607.27967](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE

The pair

Q studies mechanisms that induce AI agents to report private types truthfully and obey prescribed actions. P uses multimodal pairwise comparisons and rank aggregation to assign credit during cooperative multi-agent learning.

Connexion pursued

The initial question was whether Q's revelation and peer-scoring results could establish strategic robustness for P's credit pipeline. They cannot be applied directly: P asks an external model to compare agents' egocentric images, whereas Q scores agents' own type reports and checks joint report-and-action deviations [\[ledger 1, 2\]](#). The more precise connexion concerns whether P's rank-derived reward shaping changes strategic incentives, especially when agents leave and re-enter an active team [\[ledger 3, 5\]](#).

What was found

P defines a potential from rank-aggregated scores and shapes each transition by $F_t^i = \gamma\psi_{t+1}^i - \psi_t^i$, with zero terminal potential. If every transition is credited to agent i , the discounted shaping return telescopes:

$$\sum_{t=0}^{T-1} \gamma^t F_t^i = -\psi_0^i + \gamma^T \psi_T^i = -\psi_0^i.$$

For a fixed initial potential, this changes no unilateral-deviation payoff difference, even when actions affect intermediate images and ranks. It therefore cannot, by itself, supply Q-style incentives for honesty or obedience. It may still change finite-training dynamics, which P studies empirically [\[ledger 3, 4, 9\]](#).

P defines scores for active agents, skips comparisons when fewer than two are active, and allows battery-driven exit and re-entry. The paper does not specify how its variable-length

potential is extended to inactive agents or which transitions receive shaping rewards [ledger 5, 6]. Conditional on a fixed-coordinate potential and an indicator m_t^i for whether transition t is credited, the exact accounting identity is

$$\sum_{t=0}^{T-1} \gamma^t m_t^i (\gamma \psi_{t+1}^i - \psi_t^i) = -m_0^i \psi_0^i + \sum_{t=1}^{T-1} \gamma^t (m_{t-1}^i - m_t^i) \psi_t^i,$$

again using $\psi_T^i = 0$. Thus participation boundaries can leave path-dependent terms when rewards are credited only on selected transitions [ledger 6, 9]. For example, with $T = 2$, $m_0^i = 0$, $m_1^i = 1$, $\psi_0^i = \psi_2^i = 0$, and $\psi_1^i = q > 0$, the entering transition's positive term is omitted and the credited shaping return is $-\gamma q$. This establishes a boundary effect, not a profitable strategy for inflating rank [ledger 12, 14, 16].

P's statistical robustness claim also has a narrower scope. It concerns estimation of a stable Bradley–Terry preference vector from comparisons. In a two-agent example, if manipulable observations systematically change the probability of a comparison win to p , additional queries make the fitted score difference approach $\text{logit}(p)$, rather than the score associated with contribution-grounded probability p^* . More queries reduce sampling variation around the changed target; they do not remove that systematic change. Whether P's agents can alter visible cues independently of useful contribution remains untested [ledger 7].

Objections and resolution

The peers rejected the broad claim that P already contains strategic reporting or that Q certifies its robustness [ledger 1, 2]. They also corrected an exit example whose generic payment mask did not match the natural convention of zero potential while inactive; the entry example above is the supported conditional witness [ledger 10, 12, 14]. No implementation defect in P has been established, because its inactive-transition bookkeeping is unspecified [ledger 14, 16]. Q's peer-scoring construction requires distinct beliefs about co-player types and strong report-contingent rewards, neither of which P supplies [ledger 8, 9, 10].

Sources and remaining question

No sources outside Q and P were used for this account; the arguments above come from the papers and the numbered ledger. The material is thin on implementation and behavior: the actual reward ledger, the controllability of LMM-visible cues or battery timing, and the feasibility of a bounded report-contingent incentive mechanism remain open [ledger 16].

The smallest next step is to inspect P's reward bookkeeping at exit and re-entry, embed its potential in fixed agent coordinates, and check the discounted shaping sum on one recorded trajectory. That would determine whether the participation-boundary term arises in its implementation. A subsequent controlled comparison could test whether changing camera views or participation alters ranks without improving contribution [ledger 8, 9, 16].